Shivam Singh

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EDUCATION

B.Tech - Electronics and Communication Engineering — Uttar Pradesh, India | 2017-2021
Raj Kumar Goel Institute Of Technology, Ghaziabad — CGPA: 8.07/10

MS - Control and Optimization — Chennai, India | 2022
Indian Institute of Technology Madras — CGPA: 8.5/10

MS - ECE — Hyderabad, India | 2023-2025
International Institute of Information Technology, Hyderabad — CGPA: 8.33/10

RESEARCH EXPERIENCE

AMD INDIA PRIVATE LIMITED | Research Fellow — AI Models & Agents Team | Jun 2025 - July 2026
Supervisors: Saptarshi Majumder, Pratik Prabhanjan Brahma

- Pause and Think: Video-Grounded Assistive Action Suggestion: *2025*
  - Proposed a reasoning-centric framework for video-grounded assistive action suggestion by introducing *Pause-and-Think-T* (∼**10k** structured reasoning QA pairs) with explicit reasoning supervision and *Pause-and-Think-B*, a **300-sample** benchmark for grounded contextual understanding and goal-oriented planning across Epic-Kitchens, Assembly101, and Ego4D.
  - Designed a scalable data generation pipeline comprising annotation refinement with GPT-OSS-120B, goal-oriented clip segmentation (5–20s), and structured <think>/<answer> QA generation using Qwen3-VL-235B with majority voting and self-consistency verification.
  - Fine-tuned a compact **4B** transformer-based VLM using LLaMA-Factory with DeepSpeed on 8×AMD Instinct MI325 GPUs and served with vLLM, achieving **58.0%** accuracy—matching Qwen3-VL-235B (**58.9%**, **59× larger**) and GPT-5.2 on scene understanding, outperforming GPT-4o, and demonstrating strong zero-shot generalization on EgoThink and TempCompass without benchmark-specific training.
  - Deployed the optimized model on an **AMD RyzenTM AI** platform for real-time on-device inference, demonstrating efficient edge deployment of reasoning-capable Vision-Language Models.
  - **First Author**: Led the research, algorithm design, dataset creation, model training, evaluation, deployment, and manuscript preparation. [Code | Paper]
- Spatial Video Understanding from Raw RGB Videos: *2026*
  - Proposed *SlotMem*, a memory-augmented transformer framework for object tracking, spatial understanding, and long-horizon video reasoning from raw monocular RGB videos.
  - Developed a scalable pipeline combining Depth-Anything-3, YOLO-World, and Vision-Language Models to reconstruct 3D scenes, estimate camera trajectories, and generate supervision for spatial, temporal, and goal-oriented reasoning.
  - Trained SlotMem using our automatically generated supervision together with diverse videos from VLM-3R and VICA-322K, achieving **state-of-the-art** performance on OST-Bench (**44.4%**) and outperforming all prior baselines.

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY - HYDERABAD | M.S. Researcher
Robotics Research Centre, Hyderabad | Jan 2023 - May 2025
Direct Supervisor: Prof. Madhava Krishna
Collaborators: Dr. Mohan Sridharan, Dr. Krishna Murthy Jatavallabhula, Dr. Brojeshwar Bhowmick

- Task Anticipation using LLMs (*IEEE ICRA 2024*) *2023*
  Co-author & Project Lead: Developed an intelligent household agent that performs long-horizon planning by anticipating user actions through learning task execution patterns from user behavior and preferences, leveraging Large Language Models (LLMs) and Planning Domain Definition Language (PDDL)-based task and action representations.[Website | Paper]

- Task Anticipation for Multi-Agent Systems (*ICRA & RSS 2024 Workshops*) *2024*
  Project Lead: Extended LLM-based task anticipation to long-horizon planning in multi-agent settings, enabling robotic agents to anticipate domain-specific tasks, adapt to unexpected changes in human behavior, and handle environmental uncertainties for effective human–robot collaboration. [Website | Paper]
- LLM & Knowledge Graphs for Task Decomposition (*IEEE ICRA 2025*) *2025*
  Project Lead: Designed a framework that integrates Large Language Models (LLMs), Knowledge Graphs (KGs), and Human-in-the-Loop (HITL) feedback to enable embodied agents to perform long-horizon planning for unseen tasks and progressively adapt to new environments. [Website | Paper]
- Hybrid Planning with LLMs & Probabilistic Models (*ECMR 2025*) *2025*
  Mentor: Guided framework design and ablation experiments for task planning under uncertainty with probabilistic effects in multi-agent settings. [Paper]

PUBLICATIONS

- Anticipate & Act: Integrating LLMs and Classical Planning for Efficient Task Execution in Household Environments: R Arora*, **Shivam Singh***, K Swaminathan, A Datta, S Banerjee, B Bhowmick, KM Jatavallabhula, M Sridharan, M Krishna.
  Accepted at IEEE International Conference on Robotics and Automation (ICRA) 2024, Yokohama, Japan.
- AdaptBot: Combining LLM with Knowledge Graphs and Human Input for Generic-to-Specific Task Decomposition and Knowledge Refinement: **Shivam Singh**, K Swaminathan, N Dash, R Singh, S Banerjee, M Sridharan, M Krishna.
  Accepted at IEEE International Conference on Robotics and Automation (ICRA) 2025, Atlanta, US.
- Anticipate, Adapt, Act: A Hybrid Framework for Task Planning: N Dash, A Kaura, **Shivam Singh**, R Singh, S Banerjee, M Sridharan, M Krishna.
  Accepted at IEEE European Conference on Mobile Robots (ECMR) 2025.
- Pause and Think: A Dataset and Benchmark for Video-Grounded Assistive Action Suggestion: **Shivam Singh**, S Majumder, P Prabhanjan, Z Liu, E Barsoum.
  Accepted at IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2026. Work done at AMD (Advanced Micro Devices, Inc.).

PATENTS

- Systems and Methods for Data-driven Task Anticipation and Knowledge-driven Planning for Human-Robot Collaboration: **Shivam Singh**, K Swaminathan, R Arora, R Singh, A Datta, D Das, S Banerjee, M Sridharan, and M Krishna.
  US Patent App. 19/089,689, published November 6, 2025.

WORKSHOPS

- DaTAPlan: Data-driven Task Anticipation and Knowledge-driven Planning for Human-robot Collaboration: **Shivam Singh***, K Swaminathan*, R Arora*, R Singh, A Datta, D Das, S Banerjee, M Sridharan, and M Krishna.
  Accepted in ICRA 2024 Workshop on Cooking Robotics: Perception and motion planning, Yokohama, Japan.
- Anticipate & Collab: Data-driven Task Anticipation and Knowledge-driven Planning for Human-robot Collaboration: **Shivam Singh***, K Swaminathan*, R Arora*, D Das, S Banerjee, M Sridharan, M Krishna.
  Accepted in RSS2024 Workshop on Task Specification for General-Purpose Intelligent Robots, Delft, Netherlands

PROJECTS

- Paper implemetation: A Point Set Generation Network for 3D Object Reconstruction from a Single Image *2023*
  Pytorch, Python
  - Developed an architecture for generating 3D point clouds from single 2D images, outperforming previous methods in single-view 3D reconstruction.

- Paper implemetation: *MonoLayout*: Amodal scene layout from a single image *2024*
  Pytorch, *Python*
  - Implemented a deep learning model to predict road and vehicle layouts in bird's eye view from a single image.
  - Hallucinated occluded regions using adversarial learning for enhanced scene understanding.
- Paper implemetation: Masked-attention Mask Transformer for Universal Image Segmentation *2024*
  Pytorch, *Python*
  - Developed a universal image segmentation architecture capable of addressing any segmentation task (panoptic, instance, or semantic).
  - Utilized masked attention within the Transformer decoder to focus on localized features, significantly enhancing the model's performance on complex segmentation tasks.

WORK EXPERIENCE

ALSOENERGY – INDIA | Technical Support Engineer April 2021 | Dec 2021

- Delivered expert troubleshooting and step-by-step diagnostic support for clean energy management platforms (e.g., PowerTrack), collaborating with commercial solar clients to efficiently resolve complex issues regarding system performance, data visualization, and third-party hardware integration.

TECHNICAL SKILLS

Research Areas: Large Language Models (LLMs), Vision-Language Models (VLMs), Agentic AI, Multimodal Learning, Knowledge Representation, Temporal Reasoning, Memory-Augmented Models, Embodied AI
Frameworks & Infrastructure: PyTorch, Hugging Face Transformers, LLaMA-Factory, DeepSpeed, vLLM, ROCm, Docker, Distributed Training, OpenCV, Open3D
Programming Languages: Python, C++, C, PDDL, RDDL
Robotics & Simulation: ROS, Gazebo, AI2-THOR